

# Management of ACUTE HYPERKALAEMIA in adults

(see reverse for protocol)

First name:

Surname:

NHS number:

DOB:

Consultant:

ENTER DETAILS OF KNOWN ALLERGIES / SENSITIVITIES AND REACTION OR WRITE NIL KNOWN

THIS SECTION MUST BE COMPLETE BEFORE MEDICINES ARE GIVEN

| Initial investigations | Result | Date and time |
|------------------------|--------|---------------|
| Potassium              |        |               |
| Blood glucose          |        |               |

Protect the heart (if ECG changes present)

Within first 15 minutes

| Date | Time | Route | Drug                  | Dose  | Time to run                          | Prescriber signature & contact no | Time given | Signature(s)    |            | Asset no. of pump (if used) |
|------|------|-------|-----------------------|-------|--------------------------------------|-----------------------------------|------------|-----------------|------------|-----------------------------|
|      |      | IV    | calcium gluconate 10% | 30 mL | 10 minutes via large peripheral vein |                                   |            | Administered by | Checked by |                             |
|      |      | IV    | calcium gluconate 10% | 30 mL | 10 minutes via large peripheral vein |                                   |            |                 |            |                             |

Shift potassium into cells

Within 30-60 minutes

| Date | Time | Route | Fluid       | Volume (mL) | Additive                   | Dose    | Infusion rate | Prescriber Signature & contact no | Time given | Signature(s)    |            | Asset no. of pump (if used) |
|------|------|-------|-------------|-------------|----------------------------|---------|---------------|-----------------------------------|------------|-----------------|------------|-----------------------------|
|      |      | IV    | glucose 50% | 50mL        | soluble insulin (Actrapid) | 6 units | 15 minutes    |                                   |            | Administered by | Checked by |                             |

| Date | Time | Route | Drug       | Dose | Prescriber signature & contact no | Time started | Given by | Repeat after 6 hours using ePMA if needed |
|------|------|-------|------------|------|-----------------------------------|--------------|----------|-------------------------------------------|
|      |      | Neb   | salbutamol |      |                                   |              |          |                                           |

Remove K<sup>+</sup> from the body (prescribe if K<sup>+</sup>≥6.5 & consider prescribing if K<sup>+</sup>≥ 6.0)

Review daily and STOP if K<sup>+</sup> ≤ 5.0

| Date | Time | Route | Drug                       | Dose     | Prescriber signature & contact no | Administered by/time | Prescribe a stat dose on chart and continue TDS using ePMA for up to 72 hours. Review dose daily and stop when K <sup>+</sup> ≤ 5.0 |
|------|------|-------|----------------------------|----------|-----------------------------------|----------------------|-------------------------------------------------------------------------------------------------------------------------------------|
|      |      | Oral  | sodium zirconium (Lokelma) | 10 grams |                                   |                      |                                                                                                                                     |

Additional prescriptions as per protocol

| Date | Time | Route | Infusion fluid | Volume | Additive |              |          | Infusion rate | Prescriber signature & contact no | Time given | Signature(s)    |            | Asset no. of pump (if used) |
|------|------|-------|----------------|--------|----------|--------------|----------|---------------|-----------------------------------|------------|-----------------|------------|-----------------------------|
|      |      |       |                |        | Drug     | Dose/ volume | Batch no |               |                                   |            | Administered by | Checked by |                             |
|      |      |       |                |        |          |              |          |               |                                   |            |                 |            |                             |
|      |      |       |                |        |          |              |          |               |                                   |            |                 |            |                             |

Monitoring

| Post-treatment time | Glucose | Potassium | Results seen by | Action taken |
|---------------------|---------|-----------|-----------------|--------------|
| 30 mins             |         |           |                 |              |
| 1 hour              |         |           |                 |              |
| 90 mins             |         |           |                 |              |
| 2 hours             |         |           |                 |              |

Monitoring

| Post-treatment time | Glucose | Potassium | Results seen by | Action taken |
|---------------------|---------|-----------|-----------------|--------------|
| 3 hours             |         |           |                 |              |
| 4 hours             |         |           |                 |              |
| 5 hours             |         |           |                 |              |
| 6 hours             |         |           |                 |              |

- Perform 12-lead ECG and monitor cardiac rhythm if serum potassium  $\geq 6.0$  mmol/L
- Check serum calcium and bicarbonate
- If  $K^+ \geq 7.0$  mmol/L, do not delay administering calcium gluconate whilst awaiting ECG
- Give empirical treatment for arrhythmia if hyperkalaemia suspected
- Exclude pseudohyperkalaemia
- Check if  $K^+$  level is chronically raised and within satisfactory range for patient (i.e. in patients with known CKD)

### MILD $K^+ 5.5 - 5.9$ mmol/L

Consider cause and need for treatment

### MODERATE $K^+ 6.0 - 6.4$ mmol/L

Management guided by clinical condition, ECG and rate of rise

### SEVERE $K^+ \geq 6.5$ mmol/L

Emergency treatment indicated

First 15-30 minutes

Within 30-60 minutes

#### Are ECG changes present?

- Peaked T waves
- Flat/absent P waves
- VT
- Broad QRS
- Bradycardia
- Sine wave

**Yes**  
Cardiac monitoring/resus  
Consider referral to outreach

#### IV Calcium Gluconate 10% (protect the heart)

- Administer 30 mLs over 10 minutes via large bore IV access
- Consider administering over 30 minutes if concerns regarding digoxin toxicity
- Consider further dose if adverse ECG changes remain after 5 minutes

**No**

#### Insulin/Glucose IV infusion (shift $K^+$ into cells)

- 6 units soluble insulin (Actrapid) in 50 mL 50% (25 grams) glucose over 15 minutes
- Ensure CBG checked prior to administering
- If pre-treatment blood glucose  $<7.0$  mmol/L, consider following with 250 mL of 10% glucose infusion at 50 mL/hour for 5 hours to reduce the risk of hypoglycaemia

**Consider**

**Recommended**

#### Nebulised Salbutamol (shift $K^+$ into cells)

10-20 mg (consider giving 10 mg in patients with ischaemic heart disease). Avoid if tachycardic

#### Sodium Zirconium/Lokelma (remove $K^+$ from body)

10 grams TDS for 72 hours in life threatening hyperkalaemia

#### Consider adjunctive treatments (see below)

- Discuss haemodynamically unstable patients or those with multi-organ failure with senior and escalate to outreach as appropriate
- Discuss dialysis/transplant patients with renal team
- If refractory hyperkalaemia/cause unclear or severe renal impairment, speak to renal team

#### 1.26% Sodium Bicarbonate (shift $K^+$ into cells)

If volume deplete and acidotic, seek senior help (do not give through same cannula as calcium)

#### Furosemide bolus (remove $K^+$ from body)

If overloaded and not anuric

#### Dialysis (remove $K^+$ from body)

Contact renal team if no urine output/severe renal failure/refractory hyperkalaemia

#### Ongoing monitoring

##### Potassium

- If severe hyperkalaemia, check at 1, 2, 4 and 6 hours then again at 24 hours
- If moderate hyperkalaemia, consider checking at 1-2 hours, 4-6 hours and again at 24 hours
- In mild hyperkalaemia, check potassium daily

##### Capillary blood glucose monitoring

- Monitor capillary glucose in all patients with hyperkalaemia
- In those who have received insulin/glucose treatment, check glucose at baseline then at 30 mins, 60 mins, 90 mins, 120 mins. Continue hourly for 4 more hours

#### Establish cause of hyperkalaemia/prevent further rise or recurrence

In all cases:

- Consider the underlying cause, e.g. AKI, CKD, constipation, tissue trauma, blood transfusion, metabolic acidosis
- Review medications/diet to prevent further rise or recurrence